Programme title:	OCC:SE Y2
Module title:	**SPM622**
Lesson Topic:	**Problem Project Submission**
Date:	01 OCTOBER 2026
Student Name	
Class Activity	

Research Group Dissertation & Presentation

Project Selection and Approval

Your Task

Each group must select a **real-world problem** that will form the basis of your **Research Group Dissertation and Presentation**.

Your chosen project must identify a genuine problem experienced by **individuals, businesses, organizations, or communities** and explore how **software engineering** can be used to address or improve the problem.

Submission Deadline: Friday, 2 October 2026

Each group must submit the following to the lecturer for **approval** by the deadline:

- Proposed project title
- Short description of the problem
- Proposed software solution

Important: Do not begin the full project until your proposed problem has been approved by the lecturer.

Guidelines for Choosing Your Problem Project

1. Choose a Real Problem

The problem must exist in the real world. For example, it could be a problem experienced at:

- A local business
- A school
- A clinic
- An NGO
- A municipality
- A community

Where possible, speak to real people who are affected by the problem to understand their experiences.

2. Be Specific

Narrow the problem down to a clear and manageable issue.

Example:

❌ *Healthcare problems*

✅ *Long waiting times at a clinic's appointment desk*

3. Make Sure Software Can Help

Your proposed solution must involve **software engineering** in some meaningful way. This could include:

- A web or mobile application
- An information management system
- A database system
- Process automation
- An API
- A notification or tracking system
- Another appropriate software-based solution

4. Keep the Project Achievable

Your group will consist of **three or four members**, and you will have limited time to complete the research and presentation.

Choose a project that is realistic enough for your group to **research, analyze, design, and present properly**.

5. Make Sure the Problem Is Researchable

You should be able to find sufficient evidence to support your research. Consider whether you can access:

- Research articles
- Existing software solutions
- Case studies
- Statistics or other relevant data
- User feedback
- Interviews or observations

6. Show the Value of Your Solution

Your group should be able to clearly explain:

- Who is affected by the problem?
- Who will benefit from the proposed solution?
- Why does the problem need to be addressed?
- How could the proposed solution improve the situation?
- How will you measure whether the solution is successful?

7. Be Original

Each group must choose a **different problem**.

Avoid simply copying an existing application or system. If a similar solution already exists, identify its **limitations or weaknesses** and explain how your proposed solution could address them or provide a new approach.

8. Consider Ethics and Privacy

Your project must consider ethical and privacy requirements.

Do **not** collect or use real personal information without appropriate permission. Follow relevant data-protection principles, including the **Protection of Personal Information Act (POPIA)** where applicable.

9. Choose a Problem Your Group Cares About

Choose a topic that interests your group. Genuine interest and motivation can make your research and presentation stronger.

What to Submit for Approval

Each group must submit the following information:

1. **Group Number**
2. **Names of All Group Members**
3. **Proposed Project Title**
4. **Description of the Real-World Problem**
    - Who is affected?
    - Where does the problem occur?
    - Why is the problem important?
5. **Proposed Software Solution**
    - Describe your proposed solution
6. **Why Your Group Chose This Problem**
    - Briefly explain why your group is interested in investigating this problem.

Tips for Getting Started

1. Meet as a group during this week.
2. Brainstorm 3–5 possible real-world problems.

3. Consider the scope, available evidence, feasibility, and your group's interests.
4. Shortlist your ideas.
5. Agree on one main project and one backup option.
6. Make sure all group members agree on the selected project.
7. Submit your chosen project to the lecturer for approval by Friday, 2 October 2026.

Remember: Your project must be based on a real-world problem, be researchable and achievable, and demonstrate how software engineering can contribute to solving the problem.